Mid vowel alternation in Menominee: A substance-free search account*

David Ta-Chun Shen

A substance-free logical analysis is given concerning the mid vowel alternation in Menominee, one of the frequently mentioned cases in the studies of vowel harmony. The analysis consists of only rules, which are composed of Search (identification) and Change (modification). Three ordered rules, each of which is more general than the previous one, are proposed to deal with the phenomenon that the pertinent segments i, u, and æ are not a natural class in the given language (due to the existence of a). Should this proposal be valid, it testifies the capacity of Search and Change. Moreover, it is worthwhile to test the method portrayed with other language phenomena whose condition is similar to that of Menominee.

keywords: Menominee, natural class, Search, Unify

1 Introduction

A substance-free (Hale & Reiss 2000 et seq.) analysis of the mid vowel alternation in Menominee (Bloomfield 1962; an Algonquian language [Mithun 1999: 333] of the Algic family [ibid.: 327]) will be proposed.[1] The approach taken means that no functionalist explanations are available, like markedness/typology tendency, ease of articulation/perception consideration, diachronic episodes, etc. The unavailability does not mean that those factors do not matter; instead, they should be studied on their own.[2] Phonology, as a discipline, should thus be informed accordingly, but should not include them as part of the basic vocabulary for study. The substance-free approach also indicates that the kind of Search which I employ for the analysis is what has been argued for in Dabbous et al. (2022) and Mailhot & Reiss (2007) (cf. Nevins 2010).

The general pattern of Menominee mid vowel alternation is as follows: some[3] non-low vowels (squared in (1)) have two realizations: the high version (1a, c, e) appears if followed anywhere by a high vowel (red in (1));[4] otherwise, the mid version (1b, d, f) shows up.[5]

---

* I would like to thank the organizers and the audiences of the following events for all kinds of help which I received: the Twelfth North American Phonology Conference and the first session of Algonquian Morpho-Phonology Circle. Next, I appreciate the guidance of the editors and the referees and the facilitation of the typesetters. Naturally, the mistakes are mine. Lastly, I am grateful to Charles Reiss and Tomasina Wu.

[1] Due to the employment of some basic set-theoretic concepts, the term "Logical Phonology" may from time to time appear along with Substance-Free Phonology.

[2] For more on diachrony, please refer to Hale & Kissock (2026).

[3] There are free variation and exception to the alternation in question (Bloomfield 1962: 97).

[4] Bloomfield (1962: 96) mentions that "the semivowels y, w after a nonsyllabic" also give rise to the high realization of the alternating vowels. This conditioning factor will not be further discussed below, though Milligan (2000: 248) considers that the post-nonsyllabic glides are part of syllable nucleus.

[5] Bloomfield (1962: 96) considers that the alternation happens to the long vowels e and o and the short vowel o before glottal stop. Milligan (2000: 252) considers that the short mid vowels not followed by a glottal stop undergo the alternation in question and that the effect of the alternation is masked as the phonetic differences among the

(1) alternation of non-low vowels in Menominee (Oxford 2016: 3)

a. pi:tukuaq ‘when they bring it’ b. pi:tok ‘when he brings it’

c. mu:skamit ‘if he emerges’ d. mo:skamow ‘he emerges’

e. ki:wianæ:w ‘he takes him home’ f. ke:wæ:w ‘he goes home’

g. ne:cenæ:niw ‘my fellow man’

The predicted high version is not realized when a low front vowel (green in (1)) comes between the alternating vowel and the high vowel (1g). Moreover, the relative position matters: if the low front vowel comes after the high vowel (1e), this peculiarity is not observed and the predicted high version does appear.

Although the alternating vowels (between high and mid) can be characterized as the natural class [+syllabic, −low], the same thing cannot be said about the conditioning vowels i, u, and æ: the non-mid vowel grouping wrongly includes the low back vowel a; see (1c). The situation that i, u, and æ cannot form a natural class in Menominee is then the challenge which any analysis must face. One solution is to underspecify the low back vowel, for example, in Archangeli & Pulleyblank (1994: 375) of Grounded Phonology and Oxford (2016: 11) of the Contrastive Hierarchy. Theoretical assumptions aside, I, within my limited knowledge, seem not to be able to find other phonological process(es) in Menominee, besides the alternation which we are focusing on here, to independently support the decision to underspecify a. Therefore, the a-underspecification will not be taken for the following analysis. Another solution is to consider the relevance of some other feature, for instance, ATR (Archangeli & Pulleyblank 1994: 375; Milligan 2000: 247). It is, however, unsettled whether such a feature can be independently argued for; see Kimper’s (2011: 113) and Oxford’s (2016: 7) disagreements about ATR. So, for the following analysis, I will assume that the pertinent features are Syllabic, High, Low, Back, and Round. This in turn indicates that I favor a height-based analysis in general. Meanwhile, I am about to do so with as few theoretical assumptions as possible. If the alternation which we are discussing is indeed “a severe test for any theory of neutrality effects” (Archangeli & Pulleyblank 1994: 374), the following analysis should be viewed as one of the strong candidates to tackle such a challenge.

Below I will give an empirical demonstration on how Search and Change (Dabbous et al. 2022) work through the alternation observed in (1). The other purpose is to echo with one of the points argued in Gorman and Reiss (2026): phonological processes, assimilation or not, are uniformly broken down into Search and Change.

short front non-low vowels not followed by a glottal stop are neutralized. The irrelevance of vowel length is adopted by, say, Oxford (2016: 4) and Walker (2018: 6), and I will assume this stance as well.

2 Analysis

I use set intersection, collecting the common members from a group of sets, to determine the underlying representation of the alternating vowels. Since the +high of the high realization (i, u) and the –high of the mid realization (e, o) are not identical, these two valued features are not in the end result of set intersection. I will then assume that these alternating vowels are not specified for the feature High but are specified for the other four features (Syllabic, Low, Back, and Round) and that they are shorthanded as the capitals E and O. (All the other non-alternating vowels, of course, are fully specified in terms of these five features.)[6] It is important to immediately emphasize that there is no causal relationship between underspecification and Search (to be explained); the metaphor of lack-of-to-desire-for is not relevant at all. Furthermore, alternation does not guarantee underspecification—it is an analyst who weighs between representation and rule.

A rule consists of Search and Change (Dabbous et al. 2022): the former identifies the presence of a given configuration, and the latter then modifies the configuration in a certain way. Search contains the three parameters: initiator (INR), direction (DIR), and terminator (TRM).[7]

(2) basics of Search
INR: [$\pm F_1$, $\pm F_2$…], INR, TRM
DIR: left(ward), right(ward)
TRM: [$\pm G_1$, $\pm G_2$…], INR, TRM

All the segments whose feature specification is a superset of what is specified in INR will initiate their own search from themselves. They will look either leftward or rightward (DIR) for the *first* segment whose feature specification is a superset of what is specified in TRM and stop. They will also terminate their own search if the end of a string is reached, which means that no TRM is found. That the TRM found is the earliest encountered can make a difference if some other properties of the TRM are checked in order to determine whether Change takes

6 The feature matrices thus assumed for the current study are as follows:

i: {+syllabic, +high, −low, −back, −round}	u: {+syllabic, +high, −low, +back, +round}
E: {+syllabic, −low, −back, −round}	O: {+syllabic, −low, +back, +round}
e: {+syllabic, −high, −low, −back, −round}	o: {+syllabic, −high, −low, +back, +round}
æ: {+syllabic, −high, +low, −back, −round}	a: {+syllabic, −high, +low, +back, −round}

7 Following Bale et al. (2014), a pair of braces ({…}) is reserved for a set, and a pair of square brackets ([…]) is used for a natural class (i.e. supersets of a set). (If every member of set A is also a member of set B, A is a subset of B and B is a superset of A.) So, a traditional rule should have the following parenthetic representation: $^{A}[\ldots]^{A} \rightarrow {}^{B}\{\ldots\}^{B} \,/\, {}^{C}[\ldots]^{C}$ ___ $^{D}[\ldots]^{D}$. The input (A) and the possible environment (C, D) all refer to natural classes, but the output (B) only refers to a set.

place or not.[8] Notice that Search only identifies two points, that is, INR and TRM. And we learn from the phonological literature that there are times when three and maybe more points, if we think in terms of Search, are needed within a single rule. Not only we need more points, but we also need these points to be connected in some way. For this reason, an initiator which participated in an earlier search (displayed as the small capital INR) and a terminator which was found by an earlier search (displayed as the small capital TRM) are included in the INR and TRM parameters in (2). We are going to see an instance of how these previously accessed points work in tandem in section 2.3.

Change, by default, happens to those INRs which successfully found their TRM, unless it is stated otherwise. The two intrasegmental changes mentioned in Bale et al. (2014: 244–247) are subtraction and unification, and they are defined as follows:

(3) intrasegmental changes
a. subtraction
$A - B = \{\alpha X: \alpha X \in A \wedge \alpha X \notin B\}$
b. unification

$$A \sqcup B = \begin{cases} A, & \text{if } [\exists F_i\, (F_i \in \{F_1,\ldots,F_n\})] \wedge (\{+F_i, -F_i\} \subseteq A \cup B); \\ A \cup B, & \text{otherwise.} \end{cases}$$

Subtraction is (set-theoretic) difference: the elements in A which are shared with B are not the members of the difference of A and B. For example, {−sonorant, +voiced} – {+voiced} results in an obstruent without voicing specification ({−sonorant}). Unification of A and B returns with A alone if there is a feature from the universal features set ($\{F_1,\ldots,F_n\}$) and the plus-valued version and the minus-valued one of this feature are both the members of the union of A and B; otherwise, unification is the same as set union, collecting all the elements of each member of a group of sets. Let's see the following example.

(4) three scenarios of unification
a. non-vacuous unification
A: {+syllabic, −low}, B: {+high}; $A \sqcup B = A \cup B$: {+syllabic, −low, +high}

[8] Or more vividly, the situation is described as the hat-taking issue:

> Suppose you are told to go out into the world, find a man with a hat, and take his hat. …But the outcome is potentially different if you are told to go out, find a *person* with a hat, and take the hat *only if that person is a man*. (Mailhot & Reiss 2007: 43)

In the former scenario, if a man with a hat can be found, the instruction-follower will always end up with a hat. In the latter scenario, the instruction-follower will only end up with a hat if the found behatted person is a man. If the found behatted person is a non-male, the instruction-follower will end up hatless.

In terms of Search and Change, this difference results from where to put the properties at issue: either as part of TRM specification or as conditions on Change. More discussion of this is in Dabbous et al. (2022).

b. vacuous unification

A': {+syllabic, −low, +high}, B: {+high}; A' ⊔ B = A' ∪ B: {+syllabic, −low, +high}

c. unification failure

A'': {+syllabic, −low, −high}, B: {+high}; A'' ⊔ B = A'': {+syllabic, −low, −high}

No feature's plus and minus versions are the members of the union of A and B (4a) and those of the union of A' and B (4b); so, the unification is the same as the union. In (4c), the set {−high, +high} is a subset of the union of A'' and B {+syllabic, −low, −high, +high}; therefore, the unification returns with A'' only.

The first of the three rules to be proposed is now given.

2.1 +High specification

This rule deals with the situation where the non-low vowel is followed by the high vowel i or u:

(5) +high spec.

Search INR: [+syllabic, −low]; DIR: right; TRM: [+syllabic, +high, −low]

Change INR ⊔ {+high}

'Unify +high with every non-low vowel followed by a high vowel.'

In (5), the INR is derived by keeping what is shared between E and O: Back (−back vs. +back) and Round (−round vs. +round) are different, so only +syllabic and –low are retained. The same mechanism (set intersection) operates on i and u for the TRM. Observe that the resulted "[+syllabic, −low]" actually contains more than just E and O, so i, u, and (derived or underlying) e and o are included as well. Even though the e/o followed by a high vowel undergoes the change of (5), unification returns the e/o unaltered—{+syllabic, −high, −low,…} ⊔ {+high} = {+syllabic, −high, −low,…}, an instance of unification failure (4c).

Those E/O's which are not followed by a high vowel are realized as e/o, which leads us to the second of the three rules needed.

2.2 General –high specification

The rule below changes the non-low vowels unconditionally into the mid ones:

(6) –high-g.

Search INR: [+syllabic, −low]

Change INR ⊔ {−high}

‘Unify –high with every non-low vowel.’

Here we see how a context-free rule is phrased in Search: the parameters for DIR and TRM are left blank and not manifested. The change effect on the non-low vowels is as follows: i) for e and o, the effect is vacuous since they are already −high; ii) for E and O, they are turned into e and o non-vacuously; iii) for i and u, which are +high, they remain the same after the unification with −high.

The current rule has to be applied after +high spec. (+high spec. ≺ −high-g.) for both rules to contribute to the overall result. With only these two rules, the E/O of the string …E/O…æ…i/u… would have the following derivation: i) after +high spec., the E/O becomes i/u as E/O are indeed followed by i/u, irrespective of the intervening low front vowel; ii) after −high-g., the derived i/u is still i/u; iii) the output then is …i/u…æ…i/u…—this, however, is *not* what happens in the Menominee language.

2.3 Specific –high specification

Recall that when both the high vowel and the low front vowel appear after the non-low vowel, the former two’s positions to each other matter. If the high vowel precedes the low front vowel (…E/O…i/u…æ…), the non-low vowel becomes high due to +high spec. (…i/u…i/u…æ…). Should the order be reversed (…E/O…æ…i/u…), it is the mid vowel that shows up (…e/o…æ…i/u…). Since we have to pay attention to the two natural classes (the high vowels i/u and the low front vowel æ), we know beforehand that some connected searches will be involved, that is, searches whose INR and/or TRM parameters contain INR and/or TRM from some previous searches. Let’s, nonetheless, first see some possible scenarios which help determine the last rule of the three.

(7) two hypothesis-forming strings, where arrows stand for searches

a. the nearer i/u wrongly ignored

…E/O…i/u…æ…i/u…

b. the farther i/u wrongly included

…i/u…E/O…æ…

The analysis depicted in (7a) is straightforward: i) let E/O look rightward for æ; ii) let the found æ look rightward for i/u. But there are chances, as shown in (7a), that the nearer i/u are incorrectly missed. The remedy analysis described in (7b) solves the (7a) problem with another problem for itself: i) let E/O look rightward for æ; ii) let the found æ look leftward for i/u. But it is possible, as shown in (7b), that the i/u found by the æ is to the left of E/O and is thus irrelevant to the alternation in question. The rule to be proposed uses the following tripartite searches to overcome the aforementioned problems: i) let E/O look rightward for i/u; ii) let the same E/O look rightward for æ; iii) let the æ found from (ii) look rightward for the i/u found from (i). Since the searches (i-ii) start from the same E/O and both look rightward, we are certain that the i/u and the æ thus found, if present, are to the same side of E/O and that the problem encountered in (7b) is avoided. The search (iii) indicates how Search indirectly performs distance comparing: if the found æ can to its right farther find the found i/u, we then can be sure that æ precedes i/u. This comparison is valid because the previous two searches set the E/O as the point of origin (of a line) and establish that E/O precedes i/u and that E/O precedes æ.[9] In this manner, the problem encountered in (7a) does not arise:

(7a') with tripartite searches

cannot to its right find the <u>previously found</u> i/u

…E/O…i/u…æ…i/u…

Although there is indeed an i/u to the right of the æ in (7a'), it is *not* the terminator which is sought. Moreover, after these three searches, the string …E/O…i/u…æ… will not be judged as the same as the string …E/O…æ…i/u…, which is a welcome result to achieve.

The above preparation when phrased formally and followed by Unify is as follows:

(8) –high-sp.

Search INR-a: [+syllabic, −low]; DIR-a: right; TRM-a: [+syllabic, +high, −low]
‘Let every non-low vowel look to its right for a high vowel.’
INR-b: INR-a; DIR-b: right; TRM-b: [+syllabic, −high, +low, −back, −round]
‘Let every INR-a look to its right for a low front unrounded vowel.’
INC-c: TRM-b; DIR-c: right; TRM-c: TRM-a
‘Let every TRM-b look to its right for a TRM-a.’

Change INR-a $\sqcup$ {−high}
‘Unify –high with the INR-a’s whose Searches a to c are all successful.’

Unlike in (5) and (6), where the prose translation is for the whole rule, each of the subparts in (8) is accompanied by a prose translation. If all the three searches are successful, the configuration …E/O…æ…i/u… is present (ignoring the other non-low vowels for they will not have the non-vacuous effect), the unification then turns the E/O of this configuration into e/o.

[9] It might help to think this in terms of numbers: e.g. $0 < 5$, $0 < 6$, and $5 < 6$, so $0 < 5 < 6$.

All the other E/O's in any other configurations will leave −high-sp. unchanged (still unspecified for the feature High). When +high spec. later applies, the E/O's will be turned into i/u as long as they are followed by a high vowel. Finally, the still remaining E/O's will become e/o after the application of −high-g. In sum, the ordering among the three rules is: −high-sp. ≺ +high spec. ≺ −high-g. From looking for the presence of the low front vowel and the high vowel, to doing so only for the high vowel, and to not finding, the rules must be applied in the above order for me (as an analyst) to postulate them and for their effects to be partially observable.

To make perspicuous what the aforementioned three rules yield, the schematic derivations depending on the existence of the high vowel or the low front one are provided. Although more segments are involved for the three rules because of the natural class [+syllabic, −low], the sole attention will be on E and O (merely for the sake of clarity with no theoretical implications intended):

(9) E/O-focused schematic derivations, where the effect of the rule is non-vacuous is not shaded

input / rule	E/O…	E/O…i/u	E/O…æ	E/O…i/u…æ	E/O…æ…i/u
−high-sp.	no i/u and æ	no æ	no i/u	not what is required	e/o
+high spec.	no i/u	i/u	no i/u	i/u	unification failure
−high-g.	e/o	unification failure	e/o	unification failure	vacuous unification
output	e/o…	i/u…i/u	e/o…æ	i/u…i/u…æ	e/o…æ…i/u

3 Conclusion

Like Leduc et al. (2024), the foregoing pages presented another demonstration of the workings of Search and Change based on a fragment of Menominee phonology. The three posited rules together provided a substance-free and logical take on the particularities of the non-low vowel alternation. Whatever the phonological processes were, be it harmony-related or not, they were the result of the interaction between the parameter setting of Search and the operation of unification, plus underspecification and rule ordering, of course.

Though the current analysis began with the non-low vowel alternation in Menominee, the method of indirect distance comparing shall be applicable in other cases, if the analysis is somehow on the right track. Among others, I speculate that the vowel harmony in Finnish could be such a case: the natural class of the harmonizing vowels inevitably includes the neutral vowels, which, mutatis mutandis, is reminiscent of what we saw earlier. The actualization of the details, nevertheless, could only be taken up later.

Different theoretical approaches have their own cores to emphasize, but it is safe to say that the common goal is to generate/derive the various surface phenomena with the fewest theoretical constructs. I hope that the above demonstration will urge us (me included) to constantly (re-)examine the necessity of what is assumed. “Kids don’t need Occam’s Razor” (Gorman and Reiss 2026), but analysts need it and need it sharp.

References

Archangeli, Diana & Pulleyblank, Douglas. 1994. *Grounded phonology*. Cambridge, Massachusetts: The MIT Press.

Bale, Alan & Papillon, Maxime & Reiss, Charles. 2014. Targeting underspecified segment: A formal analysis of feature-changing and feature-filling rules. *Lingua* 148. 240–253.

Bloomfield, Leonard. 1962. *The Menomini language*. New Haven: Yale University Press.

Dabbous, Rim & Leduc, Marjorie & Mousavi, Fatemeh & Reiss, Charles & Shen, David Ta-Chun. 2022. Satisfying long-distance relationships (without tiers): A strictly anti-local approach to phonology. Ms., Concordia University and Rutgers University.

Gorman, Kyle & Reiss, Charles. 2026. Get rich quick: Why kids don’t need Occam’s Razor. *Glossa: A Journal of General Linguistics* 11.

Hale, Mark & Kissock, Madelyn. 2026. Language change in a substance-free framework. *Glossa: A Journal of General Linguistics* 11.

Hale, Mark & Reiss, Charles. 2000. Phonology as cognition. In Burton-Roberts, Noel & Carr, Philip & Docherty, Gerard (eds.), *Phonological knowledge: Conceptual and empirical issues*, 161–184. Oxford: Oxford University Press.

Kimper, Wendell A. 2011. Competing triggers: Transparency and opacity in vowel harmony. Amherst: University of Massachusetts Amherst dissertation.

Leduc, Marjorie & Reiss, Charles & Volenec, Veno. 2024. Votic vowel harmony in Substance-Free Logical Phonology. In Ritter, Nancy A. & van der Hulst, Harry (eds.), *The Oxford handbook of vowel harmony*, 425–436. Oxford: Oxford University Press.

Mailhot, Frédéric & Reiss, Charles. 2007. Computing long-distance dependencies in vowel harmony. *Biolinguistics* 1. 28–48.

Milligan, Marianne. 2000. A new look at Menominee vowel harmony. In Nichols, John D. (ed.), *Papers of the thirty-first Algonquian conference*, 237–254. Winnipeg: University of Manitoba.

Mithun, Marianne. 1999. *The languages of Native North America*. Cambridge: Cambridge University Press.

Nevins, Andrew. 2010. *Locality in vowel harmony*. Cambridge, Massachusetts: The MIT Press.

Oxford, Will. 2016. Menominee vowel harmony revisited: A height-based underspecification account. *Toronto Working Papers in Linguistics* 35. 1–14.

Walker, Rachel. 2018. Feature identity and icy targets in Menominee vowel harmony. In Bennett, Ryan & Angeles, Andrew & Brasoveanu, Adrian & Buckley, Dhyana & Kalivoda, Nick & Kawahara, Shigeto & McGuire, Grant & Padgett, Jaye (eds.), *Hana-bana (花々): A festschrift for Junko Ito and Armin Mester*, 26 pages. Retrieved from https://escholarship.org/uc/item/3xv2s3sm